2.15 What is the smallest prime divisor of $5^{23} + 7^{17}$?

2.19★ What are the 5 smallest prime numbers greater than 1000? **Hints:** 102

3.3.2 Ying and Catherine walked to the store together to buy some pencils. Ying bought 40 pencils and Catherine bought 24. If each package of pencils sold at the store contain the same number of pencils, what is the largest possible number of pencils in a package?

Important: This method for finding the GCD comes from repeatedly using the fact that

$$\gcd(m, n) = \gcd(m - n, n),$$

for any positive integers m and n with $m > n$. The method is known as the **Euclidean algorithm** after the mathematician Euclid who lived over 2,000 years ago.

Important: Let m and n be integers such that $m = qn + r$, where $0 \leq r < n$. Then

$$\gcd(m, n) = \gcd(r, n).$$

The **extended Euclidean algorithm** applies this simplification repeatedly to determine the GCD of two integers.

3.27 What is the smallest number of marbles that could either be divided up into bags of 18 marbles or into bags of 42 marbles with no marbles leftover in each case?

3.28 What is the result when we decrease the greatest common divisor of 6432 and 132 by 8?

3.31 We can construct 5 and 10 as differences of multiples of 25 and 15:

$$\begin{aligned} 2 \cdot 25 &- 3 \cdot 15 = 5 \\ 1 \cdot 25 &- 1 \cdot 15 = 10 \end{aligned}$$

In this way, generate the next 8 smallest multiples of 5. **Hints:** 134

3.33★ How many positive integers less than 101 are multiples of either 5 or 7, but not both at once?

Definition: Numbers that can be expressed as the quotient of two integers are called **rational numbers.**

Here are some examples:

$$\frac{3}{4} \qquad \frac{19}{6} \qquad \frac{-16}{3}$$

$$\frac{7}{1} \qquad \frac{1}{7} \qquad \frac{-24}{-6}$$

Since division by 0 is undefined, rational numbers are numbers that we can write in the form $\frac{a}{b}$ where a and b are integers, but $b \neq 0$.

Some rational numbers (like $\frac{7}{1} = 7$) are integers. In fact, all integers are rational numbers since they can be expressed as a quotient between themselves and 1:

$$\dots \qquad \frac{-2}{1} = -2 \qquad \frac{-1}{1} = -1 \qquad \frac{0}{1} = 0 \qquad \frac{1}{1} = 1 \qquad \frac{2}{1} = 2 \qquad \dots$$

Problem 4.19: Janet writes the cubes of three positive integers on a piece of paper. Rob points out that each is a multiple of 18. Janet then points out that the GCD of all three perfect cubes is n. Find the smallest possible value of n.

4.39★ A group of 10 friends were discussing a large positive integer. "It can be divided by 1," said the first friend. "It can be divided by 2," said the second friend. "And by 3," said the third friend. "And by 4," added the fourth friend. This continued until everyone had made such a comment. If exactly two friends were incorrect, and those two friends said consecutive numbers, what was the least possible integer they were discussing? *(Source: MATHCOUNTS)* **Hints:** 34

4.40★ What is the smallest positive integer N such that the value $7 + 30N$ is not a prime number?

4.50★ Find the smallest positive integer greater than 1 which leaves a remainder of 1 when divided by $2, 3, \ldots, 8$, and 9. *(Source: Mandelbrot)* **Hints:** 127

4.53★ How many possible values are there for the sum $a + b + c$ if a, b, and c are positive integers and $abc = 72$? *(Source: Mandelbrot)*

4.57★ Ben, Fang, and Venetia are playing a game in which a card numbered 2, 3, 4, 5, 6, 7, or 8 is stuck to each of their foreheads, so that each player can see the other two numbers but not their own. Joey walks in and observes that the three numbers are not all different, and that the product of the three numbers is a perfect square. How many of the three players can now deduce the numbers on their foreheads? *(Source: Mandelbrot)* **Hints:** 87, 38

Problem 5.12: Find the product of all of the positive divisors of 450 that are multiples of 3.

5.4.7 Find the product of the positive divisors of 2400 that are multiples of 6.

5.26 Find the smallest positive integer that can be written as a product of two factors (each greater than 1) in exactly three different ways. For example, 70 is one such integer since $70 = 2 \cdot 35 = 5 \cdot 14 = 7 \cdot 10$. Note that the order of the factors does not matter. *(Source: Mandelbrot)* **Hints:** 110

5.31 What is the sum of all positive integers less than 100 that have exactly twelve divisors?

5.33★ Find the sum of the perfect square divisors of the smallest integer with exactly 6 perfect square divisors. **Hints:** 33

5.36★ How many positive integers have exactly three proper divisors, each of which is less than 50?

A **Mersenne prime** is a prime number of the form:

$$M_p = 2^p - 1$$

where **p is also a prime number**.

Examples:

- $p = 2 \to 2^2 - 1 = 3$ (prime)
- $p = 3 \to 2^3 - 1 = 7$ (prime)
- $p = 5 \to 2^5 - 1 = 31$ (prime)
- $p = 11 \to 2^{11} - 1 = 2047$ (not prime)

Important Note:

- If p is not prime $\to 2^p - 1$ is definitely not prime
- If p is prime $\to 2^p - 1$ **may or may not** be prime

Mersenne primes are important because they are used to find **very large primes**.

A **Fermat prime** is a number of the form:

$$F_n = 2^{2^n} + 1$$

First few Fermat numbers:

n	Fermat number
0	3
1	5
2	17
3	257
4	65537

Fermat conjectured:

> All numbers of the form
> $F_n = 2^{2^n} + 1$
> are prime.

In 1732, **Euler** showed that:

$$F_5 = 2^{32} + 1 = 4294967297$$

is **not prime**, because:

$$4294967297 = 641 \times 6700417$$

This **disproved Fermat's conjecture**.

After that, mathematicians found:

- $F_6, F_7, F_8 \ldots$ are **also not prime**
- So far, **only the first five Fermat numbers are prime**

Twin primes are pairs of prime numbers that differ by **2**.

$$(p,\ p+2)$$

Both numbers must be prime.

Examples:

- (3, 5)
- (5, 7)
- (11, 13)
- (17, 19)
- (29, 31)

The **Twin Prime Conjecture** states:

> There are **infinitely many twin primes**.

In other words:

- Twin primes keep appearing forever
- There is no largest twin prime

- The **Twin Prime Conjecture has NOT been proven**
- It is still an **open problem** in mathematics

However, there has been **major progress**:

2013 — Yitang Zhang

Mathematician **Yitang Zhang** proved:

> There are infinitely many prime pairs with difference **less than 70 million**

This was the **first breakthrough**.

Mathematicians improved the bound:

- 70,000,000 → 4,680 → 246 → **6** (current best known bound)

This means:

> There are infinitely many primes that differ by **at most 6**

But we still **cannot prove difference = 2**.

Concept	Form	Status
Mersenne primes	$2^p - 1$	Some known
Fermat primes	$2^{2^n} + 1$	Only first 5 prime
Fermat conjecture	All Fermat primes	Disproved by Euler
Twin primes	$p, p + 2$	Many known
Twin prime conjecture	Infinite twin primes	Unsolved

6.3.8 What is the product of the positive divisors of 7!?

Problem 6.11: The product of two positive three-digit palindromes is 436995. What is their sum?

6.5.5★ Find the smallest three-digit palindrome whose product with 101 is **not** a five-digit palindrome.

6.25 Find the largest integer n such that n has exactly 4 positive divisors and n divides $100!$. **Hints:** 62

6.27 How many positive cubes divide $3! \cdot 5! \cdot 7!$? *(Source: AMC)*

6.31★ For how many positive integers n less than or equal to 24 is $n!$ evenly divisible by $1 + 2 + \cdots + n$?

6.36★ Let $T(n)$ represent the number of terminal zeros of $n!$ for any positive integer n (the number of zeros starting from the rightmost digit before the first nonzero digit).

(a) For which values of n is $T(n) - T(n-1) > 0$?

6.37★ In how many consecutive zeroes does the product $115 \times 116 \times 117 \times \cdots \times 201$ end? *(Source:*

6.39

(b) Find the smallest possible GCD of a pair of six-digit palindromes. **Hints: 7**

6.41★ Show that if n is either perfect or abundant, then every positive multiple of n that is greater than n is abundant. **Hints:** 107

6.43★ There are unique integers $a_2, a_3, a_4, a_5, a_6, a_7$ such that $\frac{5}{7} = \frac{a_2}{2!} + \frac{a_3}{3!} + \frac{a_4}{4!} + \frac{a_5}{5!} + \frac{a_6}{6!} + \frac{a_7}{7!}$, where $0 \le a_i < i$ for $i = 2, 3, \ldots, 7$. Find $a_2 + a_3 + a_4 + a_5 + a_6 + a_7$. *(Source: AHSME)* **Hints:** 22, 143

Simon's Favorite Factoring
a factoring technique used to rewrite expressions of the form: xy + Ax + By
One equation and two variables

Example

Factor:

$$3x + 6 + xy + 2y$$

Step 1 — Rearrange

$$xy + 3x + 2y + 6$$

Step 2 — Group

$$x(y + 3) + 2(y + 3)$$

Step 3 — Factor

$$(x + 2)(y + 3)$$

Final Answer:

$$(x + 2)(y + 3)$$

Problem 7.3: Let x and y be positive integers such that

$$\frac{1}{x} + \frac{1}{y} = \frac{1}{7}.$$

Find the sum of all possible values of x. *(Source: MATHCOUNTS)*

Problem 7.7: Anna owns some blue marbles, green marbles, and red marbles. She has $\frac{7}{5}$ as many blue marbles as green marbles and $\frac{2}{7}$ as many green marbles as red marbles. Find the smallest possible number of marbles Anna could own.

Problem 7.8: If the 3-digit positive integer $n = ABC = AB + BA + AC + CA + BC + CB$, compute the largest possible value for n. *(Source: ARML)*

Problem 7.9: Find the sum of all four-digit positive integer palindromes. *(Source: Mandelbrot)*

7.10 How many perfect squares less than 500 are divisible by 8? *(Source: MATHCOUNTS)*

7.11 Find each two-digit number that is equal to four times the sum of its digits.

7.13 There are three unique two-digit numbers such that the sum of the squares of the first number's digits equals the sum of the squares of the second number's digits equals the sum of the squares of the third number's digits, and each of these sums is equal to the sum of the squares of the digits of 345. What is the sum of these three two-digit numbers? *(Source: MATHCOUNTS)*

7.15 John has a jar of between 500 and 600 coins that are all nickels, dimes, and quarters. The ratio of the number of quarters to the number of dimes is 8 : 5 and the ratio of the number of dimes to the number of nickels is 6 : 11. What is the total value of the money in John's jar of coins?

7.20 There are 9 six-digit numbers that have the property that all of their digits are the same (such as 111111 and 222222). Find the largest prime number that is a divisor of all of them.

7.26 How many four-digit integers are even perfect squares and also perfect cubes? **Hints:** 167